LESSON PLAN · SCRATCH (GR 6–8) INTERMEDIATE: DATA & SIMULATIONS

Lesson 6: Project: Procedural City

Module 2 | 50 minutes | Project: procedures | Grades 6–8 · ages 11–14

Objective: Plan, build and test a program that draws a city skyline from procedures with parameters and randomness.

Success criteria

- My city is built from procedures with parameters.
- It uses random arguments.
- I tested my procedures on their own.

Key vocabulary

- **procedural generation**: Creating content with code and randomness.
- **design document**: A plan of procedures and parameters.
- **unit test**: Testing one small piece on its own.

You will need

- Projector with Kitabash (lesson video, Code Lab)
- Devices: 1 per learner or pair
- Printed worksheet (optional)
- Printed project brief

Timed lesson plan (50 min)

Time	Stage	What happens
5 min	**Warm-up (unplugged)**	Skyline (5 min): roll a die for each building height and draw on squared paper.
5 min	**Watch & plan**	Play the project video; learners plan on the project brief.
25 min	**Build**	Build-along steps, then Plan building (height). Check in halfway: everyone has something working.
7 min	**Test & peer feedback**	Swap and test each other's work: two stars and a wish.
8 min	**Improve & submit**	Improve one thing; Project: Procedural City.

Unplugged warm-up

Skyline (5 min): roll a die for each building height and draw on squared paper.

Teaching the idea

Procedural generation combines parameterised procedures with randomness. Assess decomposition (design document), correct parameter use, and testing of procedures in isolation.

Analogy: A city planner with stamps of different sizes.

Live demo: Write building (height) and call it with 50, 100, 150; then in a loop with random.

Questions to ask

Ask	Listen for
What is your building's parameter?	Height.
Where does randomness come in?	The argument.
How did you test a procedure?	Called it alone with known values.

Differentiation

Support
Provide building (height); learners add the loop and one detail.

Stretch
windows (height) inside building; night/day colour schemes.

Common misconceptions

Misconception	How to fix it
Each building needs its own code	One procedure, many calls.
Random means unplanned	The rules are designed; only values vary.

Assessment

Project rubric (also on the project brief):

Criterion	1 · Beginning	2 · Developing	3 · Secure	4 · Excellent
Decomposition	None	One big script	Several focused procedures	Well-layered procedures calling procedures
Parameters	None	Defined but unused	Used correctly	Several parameters used thoughtfully
Generation	Fixed picture	Some randomness	Different city each run	Varied, designed results
Visual quality	Unclear	Basic	Attractive	Detailed and striking
Testing & reflection	None	Ran once	Procedures tested alone	Explains testing and fixes

Exit ticket
Name one procedure in your city and its parameter.

Homework
Generate five cities and screenshot (or sketch) your favourite.

Answer key — Kitabash activities

1. Plan building (height) (perform · ordering)

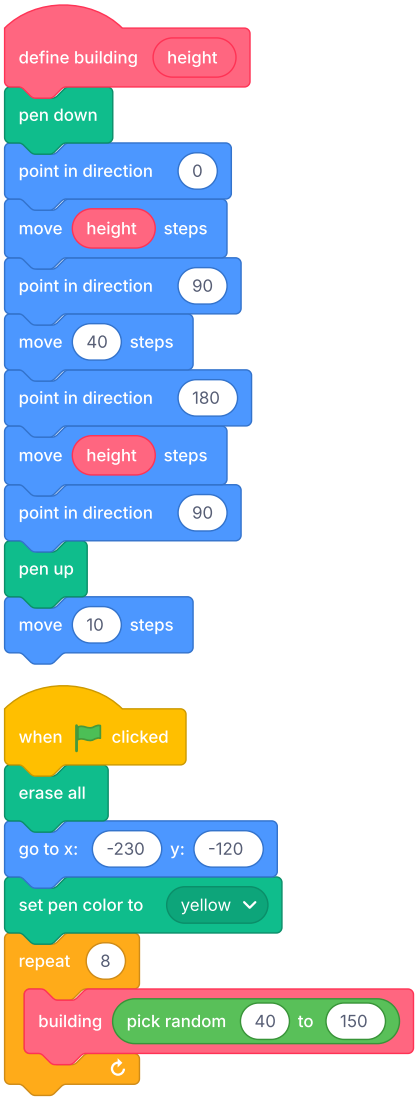
Order: Pen down at the bottom left → Move up by height → Move right 40 → Move down by height → Pen up, ready for the next building

2. Project: Procedural City (probe · code lab)

Goals checked: Procedures · Parameters · Randomness · A full skyline

Example project:

Pen



Answer key — worksheet

1. **Key words:** procedural generation = Creating content with code and randomness. · design document = A plan of procedures and parameters. · unit test = Testing one small piece on its own.

2. **Plan building (height):** 1 Pen down at the bottom left · 2 Move up by height · 3 Move right 40 · 4 Move down by height · 5 Pen up, ready for the next building
3. **My plan:** A clear plan that covers every requirement.
4. **Checklist:** All requirements met.
5. **Two stars and a wish:** Kind, specific feedback and an improvement.

Kitabash original curriculum. Scratch is a project of the Scratch Foundation in collaboration with the MIT Media Lab; this course is not affiliated with or endorsed by them.